

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2023-2024

Comprehensive Examination
(EC-3 Make-up)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : EN

No. of Pages	= 2
No. of Questions	= 7

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.

Q.1. A 2-tier application uses a 3-node application cluster and a 2-node DB cluster. The application works only when all tiers are available. The application tier is in an active-active load balanced configuration with the given nodes. But the database tier is in a cold standby mode where it takes 12 hours to bring back a passive node online. If an application node fails every 10 days and a DB node fails every 100 days, find the following:

- (a) MTTF of the application tier.
 - (b) MTTF of the database tier.
 - (c) Availability of the database tier.
 - (d) Overall availability of the 2-tier system, assuming MTTR of the application tier is negligible.
- (4 Marks)

Q.2. A company decided to perform real time analytics on its Web server logs to gain insights on website visitors, behavior, crawlers accessing the site, business insights, security issues, and more. For this, the log file entries are to be streamed to an HDFS folder in a Hadoop cluster. Streaming the logs to HDFS is to be done only when 1000 or more entries are made to the webserver log. Configure a Flume agent to implement the streaming with webserver log as the source and an HDFS folder as the sink.

(7 Marks)

Q.3. You are assigned the responsibility to transfer huge volume of data from RDBMS table to HDFS for daily analytics on the data. The time window allocated for the data transfer is fixed and the volume of the data increases day by day. Which data ingestion tool will be recommended by you. Give justifications for your selection.

(4 Marks)

Q.4. An enterprise company need to address the following challenges with Big Data Analytics:

- a. Struggling with a Java based ELT tool to manage and extract value from the growing volume and variety of data and need to unify information across federated sources. Each disparate MySQL server records around 2 million updates, a total of 50 million updates daily.

- b. Unable to relate raw data collected from system logs, sensors, or click streams with customer and line-of-business data managed in enterprise systems.
- c. Each end-to-end ELT process consumes 14 hours. This process should be completed within an hour.
- d. Some data warehousing queries on structured data are available only in SQL and there is no expertise available for transforming these SQL queries into Hadoop ecosystem tools.

Recommend suitable solutions for the above challenges by making use of Hadoop Ecosystem products/tools. Give justifications for your recommendations.

(7 Marks)

Q.5. You have a 928 MB file stored on HDFS as part of a Hadoop 2.x distribution. A data analytics program uses this file and runs in parallel across the cluster nodes.

(a) The default block size and replication factor is used in the configuration. How many total blocks including replicas will be stored in the cluster ? What are the unique HDFS block sizes you will find for the specific file?

(b) The cluster has 64 cores to speed up the processing. If the program can at best achieve 60% parallelism in the code to exploit the multiple cores and the rest of it is sequential, what is the theoretical limit on speed-up you can expect with 64 cores compared to a sequential version of the same program running on one core with the same file ? How will this limit change if you doubled the compute power to 128 cores ? You can simplify the system to assume cluster nodes and cores mean the same and we can ignore the overheads of communication etc. depending on the specific cluster configuration, scheduling etc.

(c) Suppose you could use a more scalable algorithm with 80% parallelism and a larger file as you move to a 128 core system. What would be the theoretical speed-up limit for 128 cores ?

(7 Marks)

Q.6. Spark Core uses RDD as a base data structure. Explain the resiliency aspect of an RDD with specific details of how a computation that is a sequence of transformations and actions, can survive a node / executor crash or what happens when a node runs low on memory.

(6 Marks)

Q.7. Consider an online shopping site whose transaction workload increases during festive seasons. So for this specific period of time, the resources need spike up. In order to handle this kind of situation what is the preferable choice? Is it cloud elasticity or cloud scalability give the reasons.

(5 Marks)
